Progress Report 3: Detecting Out-of-Context Image-Caption Misinformation with Multimodal Embeddings

DATA5500 Senior Design | Fall 2026 | Instructor: Weijie Pang | Student: Mohamed Mohamed | October 18, 2026

1. Tasks Completed

Task	What was built / implemented	Hours	Results and outcomes
Logistic regression classifier	Logistic regression on the full CLIP features (image, caption, element-wise product, absolute difference, cosine; 2,049 dims), standardized, regularization tuned on validation.	2.0	Test ROC-AUC 0.751 (CI 0.733 to 0.769) vs. 0.716 for the baseline; accuracy 0.682, F1 0.685 (baseline 0.637 / 0.580). Best C = 0.0003.
Model comparison with bootstrap CIs	Per-method test metrics with 95% bootstrap CIs (1,000 resamples) and a paired bootstrap of the AUC gain over the baseline on the same 2,400 test pairs.	1.5	Overall gain +0.036 (CI +0.023 to +0.050). CLIP text-image now beats chance: 0.458 to 0.567 (gain +0.109). Person, scene and CLIP text-text gains are not significant.
Image-embedding difference investigation	Re-embedded 2,641 images under 10 preprocessing variants (Hugging Face vs. OpenAI CLIP preprocessing, CPU vs. GPU, fp32 vs. fp16, resize methods, EXIF rotation).	2.0	No variant explains the gap: all give mean cosine 0.9935 to 0.9937. The gap tracks image size (0.986 under 300 px vs. 0.998 at 600 px+), so the provided embeddings likely came from different-resolution image files. My current pipeline is kept for the app.
Neural network (first draft)	PyTorch MLP (2,049 to 512 to 128 to 1, dropout, AdamW, early stopping on validation AUC).	1.5	Test ROC-AUC 0.757 (CI 0.738 to 0.774), CLIP text-image 0.589. Not significantly better than logistic regression (+0.005, CI -0.004 to +0.015); it overfits after epoch 2.
GitHub repository and documentation	Published the project to GitHub: numbered pipeline scripts, metrics, figures, trained models, docs and project PDFs, and a full README with setup, reproduction steps and results.	1.5	Repository: github.com/mohamedm-data/OOC-Misinformation-Detection. All results reproducible from 7 scripts; 13 figures.
Total		**8.5**	

Blockers, challenges, and risks

- **Risk: the 0.80 ROC-AUC target is not reached yet** (best 0.757). Logistic regression scores only 0.80 even on its training data, while the MLP reaches 0.996 on training but peaks on validation after 2 epochs. The fixed global CLIP features, not model size, look like the limit.
- **Almost all of the gain comes from the CLIP text-image split.** The other three methods stay within the baseline's confidence intervals.
- **Mislabeled image files crashed the embedding run** (23 GIFs, 33 PNGs and 1 BMP saved as .jpg; 28 grayscale or transparent images). Solved by converting every image to RGB before CLIP; the rerun covered all 2,641 images.
- **The laptop had no GitHub sign-in**, so the first pushes failed. Solved by connecting my GitHub account to the cloud workspace and pushing from there.

2. Plan for Next Week

- Train on the full balanced NewsCLIPpings training split (71,072 pairs instead of 19,200) to test whether more data closes the gap to 0.80, since the MLP overfits the current subset.
- Regularize the MLP (stronger weight decay and dropout, smaller hidden layers) and compare it again with logistic regression using the paired bootstrap.
- Error analysis on the CLIP text-image and person splits: inspect the most confident mistakes to see what the features miss.
- Write the midterm report (plan Week 7, due about October 25) with the baseline, logistic regression and MLP results by method.
- Start the Streamlit app skeleton using my own embedding pipeline, now confirmed to match the training embeddings.

Help needed

None at this time. Two choices guide next week's work:

- Since the image-embedding gap comes from the source image files rather than my code, the app keeps the current Hugging Face CLIP pipeline (mean cosine 0.994 to the provided embeddings).
- Model work stays on frozen CLIP embeddings, as designed. More training data and regularization are tried first, before any change of features.